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JEE MAIN · PHYSICS · CHAPTER-WISE PYQS

Vector Algebra

JEE Main Physics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	28 questions
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MCQ (Single Correct Answer) 28 questions

Q1. JEE Main 2026 (Online) 2nd April Morning Shift

The velocity of a particle is given as $\vec{v} = -x\hat{i} + 2y\hat{j} - z\hat{k}$ m/s. The magnitude of acceleration at point (1, 2, 4) is _____ m/s^2.

(A) $\sqrt{6}$ (B) 9 (C) $\sqrt{33}$ (D) 0

Open solution

Q2. JEE Main 2024 (Online) 5th April Morning Shift

The angle between vector $\vec{Q}$ and the resultant of $(2\vec{Q} + 2\vec{P})$ and $(2\vec{Q} - 2\vec{P})$ is :

(A) $\tan^{-1}(\mathrm{P/Q})$ (B) 0°

(C) $\tan^{-1}\frac{(2\vec{Q}-2\vec{P})}{2\vec{Q}+2\vec{P}}$ (D) $\tan^{-1}(2Q/\mathrm{P})$

Open solution

Q3. JEE Main 2024 (Online) 31st January Evening Shift

If two vectors $\vec{A}$ and $\vec{B}$ having equal magnitude R are inclined at angle θ, then

(A) $|\vec{A} + \vec{B}| = 2R\cos\left(\frac{\theta}{2}\right)$ (B) $|\vec{A} - \vec{B}| = 2R\cos\left(\frac{\theta}{2}\right)$

(C) $|\vec{A} - \vec{B}| = \sqrt{2}R\sin\left(\frac{\theta}{2}\right)$ (D) $|\vec{A} + \vec{B}| = 2R\sin\left(\frac{\theta}{2}\right)$

Open solution

Q4. JEE Main 2023 (Online) 15th April Morning Shift

A vector in $x - y$ plane makes an angle of 30° with y-axis. The magnitude of y-component of vector is $2\sqrt{3}$. The magnitude of x-component of the vector will be :

(A) $\sqrt{3}$ (B) 2 (C) 6 (D) $\frac{1}{\sqrt{3}}$

Open solution

Q5. JEE Main 2023 (Online) 11th April Evening Shift

When vector $\vec{A} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ is subtracted from vector $\overrightarrow{\mathrm{B}}$, it gives a vector equal to $2\hat{j}$. Then the magnitude of vector $\overrightarrow{\mathrm{B}}$ will be :

(A) 3 (B) $\sqrt{33}$ (C) $\sqrt{6}$ (D) $\sqrt{5}$

Open solution

Q6. JEE Main 2023 (Online) 8th April Morning Shift

Two forces having magnitude A and $\frac{A}{2}$ are perpendicular to each other. The magnitude of their resultant is:

(A) $\frac{5A}{2}$ (B) $\frac{\sqrt{5}A}{4}$ (C) $\frac{\sqrt{5}A}{2}$ (D) $\frac{\sqrt{5}A^2}{2}$

Open solution

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Q7. JEE Main 2023 (Online) 24th January Evening Shift

If two vectors $\vec{P} = \hat{i} + 2m\hat{j} + m\hat{k}$ and $\vec{Q} = 4\hat{i} - 2\hat{j} + m\hat{k}$ are perpendicular to each other. Then, the value of m will be :

(A) -1 (B) 3 (C) 1 (D) 2

Open solution

Q8. JEE Main 2022 (Online) 29th June Morning Shift

Two vectors $\vec{A}$ and $\vec{B}$ have equal magnitudes. If magnitude of $\vec{A} + \vec{B}$ is equal to two times the magnitude of $\vec{A} - \vec{B}$, then the angle between $\vec{A}$ and $\vec{B}$ will be :

(A) $\sin^{-1}\left(\frac{3}{5}\right)$ (B) $\sin^{-1}\left(\frac{1}{3}\right)$

(C) $\cos^{-1}\left(\frac{3}{5}\right)$ (D) $\cos^{-1}\left(\frac{1}{3}\right)$

Open solution

Q9. JEE Main 2022 (Online) 25th June Morning Shift

$\vec{A}$ is a vector quantity such that $|\vec{A}|$ = non-zero constant. Which of the following expression is true for $\vec{A}$?

(A) $\vec{A} . \vec{A} = 0$ (B) $\vec{A} \times \vec{A} < 0$

(C) $\vec{A} \times \vec{A} = 0$ (D) $\vec{A} \times \vec{A} > 0$

Open solution

Q10. JEE Main 2022 (Online) 25th June Morning Shift

Which of the following relations is true for two unit vectors $\hat{A}$ and $\hat{B}$ making an angle θ to each other?

(A) $|\hat{A} + \hat{B}| = |\hat{A} - \hat{B}| \tan\frac{\theta}{2}$ (B) $|\hat{A} - \hat{B}| = |\hat{A} + \hat{B}| \tan\frac{\theta}{2}$

(C) $|\hat{A} + \hat{B}| = |\hat{A} - \hat{B}| \cos\frac{\theta}{2}$ (D) $|\hat{A} - \hat{B}| = |\hat{A} + \hat{B}| \cos\frac{\theta}{2}$

Open solution

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Q11. JEE Main 2021 (Online) 31st August Evening Shift

Open solution

Statement I :

Two forces $\left(\vec{P}+\vec{Q}\right)$ and $\left(\vec{P}-\vec{Q}\right)$ where $\vec{P}\perp\vec{Q}$, when act at an angle θ_1 to each other, the magnitude of their resultant is $\sqrt{3(P^2+Q^2)}$, when they act at an angle θ_2, the magnitude of their resultant becomes $\sqrt{2(P^2+Q^2)}$. This is possible only when $\theta_1 < \theta_2$.

Statement II :

In the situation given above.

θ_1 = 60° and θ_2 = 90°

In the light of the above statements, choose the most appropriate answer from the options given below :-

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are true

(C) Statement I is true but Statement II is false

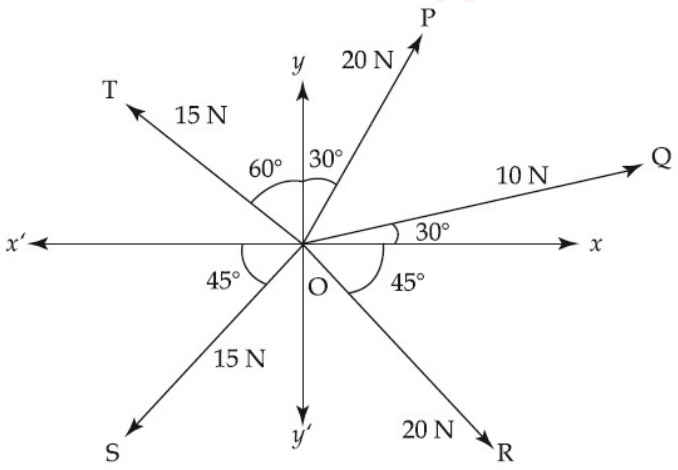
(D) Both Statement I and Statement II are false.

Q12. JEE Main 2021 (Online) 27th August Morning Shift

Open solution

The resultant of these forces $\overrightarrow{OP}, \overrightarrow{OQ}, \overrightarrow{OR}, \overrightarrow{OS}$ and $\overrightarrow{OT}$ is approximately ..... N.

[Take $\sqrt{3}=1.7$, $\sqrt{2}=1.4$ Given $\hat{i}$ and $\hat{j}$ unit vectors along x, y axis]



(A) $9.25\hat{i}+5\hat{j}$

(B) $3\hat{i}+15\hat{j}$

(C) $2.5\hat{i}-14.5\hat{j}$

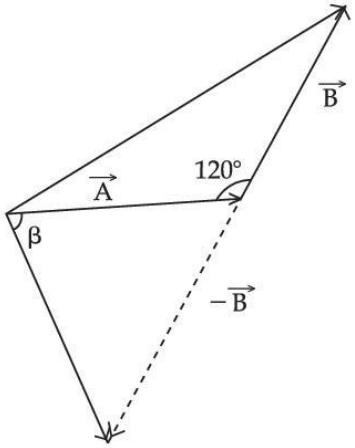
(D) $-1.5\hat{i}-15.5\hat{j}$

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Q13. JEE Main 2021 (Online) 26th August Evening Shift

The angle between vector $\left(\vec{A}\right)$ and $\left(\vec{A}-\vec{B}\right)$ is :

Open solution



(A) $\tan^{-1}\left(\frac{-\frac{B}{2}}{A-B\frac{\sqrt{3}}{2}}\right)$

(B) $\tan^{-1}\left(\frac{A}{0.7B}\right)$

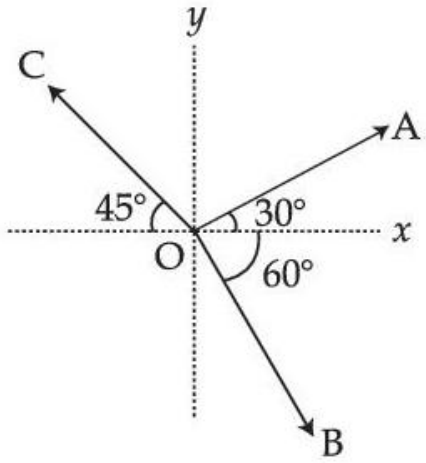
(C) $\tan^{-1}\left(\frac{\sqrt{3}B}{2A-B}\right)$

(D) $\tan^{-1}\left(\frac{B\cos\theta}{A-B\sin\theta}\right)$

Q14. JEE Main 2021 (Online) 26th August Morning Shift

The magnitude of vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OC}$ in the given figure are equal. The direction of $\overrightarrow{OA}$ + $\overrightarrow{OB}$ − $\overrightarrow{OC}$ with x-axis will be :

Open solution



(A) $\tan^{-1}\frac{(1-\sqrt{3}-\sqrt{2})}{(1+\sqrt{3}+\sqrt{2})}$

(B) $\tan^{-1}\frac{(\sqrt{3}-1+\sqrt{2})}{(1+\sqrt{3}-\sqrt{2})}$

(C) $\tan^{-1}\frac{(\sqrt{3}-1+\sqrt{2})}{(1-\sqrt{3}+\sqrt{2})}$

(D) $\tan^{-1}\frac{(1+\sqrt{3}-\sqrt{2})}{(1-\sqrt{3}-\sqrt{2})}$

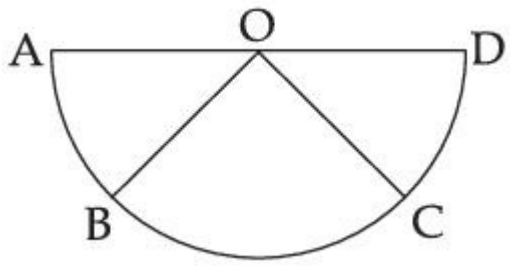
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Q15. JEE Main 2021 (Online) 27th July Morning Shift

Open solution

Assertion A : If A, B, C, D are four points on a semi-circular arc with centre at 'O' such that $\left|\overrightarrow{AB}\right| = \left|\overrightarrow{BC}\right| = \left|\overrightarrow{CD}\right|$, then $\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD} = 4\overrightarrow{AO} + \overrightarrow{OB} + \overrightarrow{OC}$

Reason R : Polygon law of vector addition yields $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{AD} = 2\overrightarrow{AO}$



In the light of the above statements, choose the most appropriate answer from the options given below :

(A) A is correct but R is not correct.

(B) A is not correct but R is correct.

(C) Both A and R are correct and R is the correct explanation of A.

(D) Both A and R are correct but R is not the correct explanation of A.

Q16. JEE Main 2021 (Online) 25th July Evening Shift

Open solution

Two vectors $\vec{X}$ and $\vec{Y}$ have equal magnitude. The magnitude of $(\vec{X} - \vec{Y})$ is n times the magnitude of $(\vec{X} + \vec{Y})$. The angle between $\vec{X}$ and $\vec{Y}$ is :

(A) $\cos^{-1}\left(\frac{-n^2-1}{n^2-1}\right)$

(B) $\cos^{-1}\left(\frac{n^2-1}{-n^2-1}\right)$

(C) $\cos^{-1}\left(\frac{n^2+1}{-n^2-1}\right)$

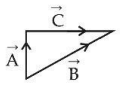
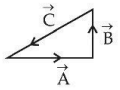
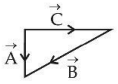
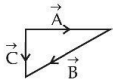
(D) $\cos^{-1}\left(\frac{n^2+1}{n^2-1}\right)$

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Q17. JEE Main 2021 (Online) 25th July Morning Shift

Open solution

Match List - I with List - II

List I	List II
(a) $\vec{C}-\vec{A}-\vec{B}=0$	(i) 
(b) $\vec{A}-\vec{C}-\vec{B}=0$	(ii) 
(c) $\vec{B}-\vec{A}-\vec{C}=0$	(iii) 
(d) $\vec{A}+\vec{B}=-\vec{C}$	(iv) 

Choose the correct answer from the options given below :

(A) (a) → (iv), (b) → (i), (c) → (iii), (d) → (ii)

(B) (a) → (iv), (b) → (iii), (c) → (i), (d) → (ii)

(C) (a) → (iii), (b) → (ii), (c) → (iv), (d) → (i)

(D) (a) → (i), (b) → (iv), (c) → (ii), (d) → (iii)

Q18. JEE Main 2021 (Online) 22th July Evening Shift

Open solution

What will be the projection of vector $\vec{A}=\hat{i}+\hat{j}+\hat{k}$ on vector $\vec{B}=\hat{i}+\hat{j}$?

(A) $\sqrt{2}(\hat{i}+\hat{j}+\hat{k})$

(B) $(\hat{i}+\hat{j})$

(C) $\sqrt{2}(\hat{i}+\hat{j})$

(D) $2(\hat{i}+\hat{j}+\hat{k})$

Q19. JEE Main 2021 (Online) 20th July Evening Shift

Open solution

Two vectors $\vec{P}$ and $\vec{Q}$ have equal magnitudes. If the magnitude of $\vec{P}+\vec{Q}$ is n times the magnitude of $\vec{P}-\vec{Q}$, then angle between $\vec{P}$ and $\vec{Q}$ is :

(A) $\sin^{-1}\left(\frac{n-1}{n+1}\right)$

(B) $\cos^{-1}\left(\frac{n-1}{n+1}\right)$

(C) $\sin^{-1}\left(\frac{n^2-1}{n^2+1}\right)$

(D) $\cos^{-1}\left(\frac{n^2-1}{n^2+1}\right)$

Q20. JEE Main 2021 (Online) 20th July Morning Shift

Open solution

If $\vec{A}$ and $\vec{B}$ are two vectors satisfying the relation $\vec{A}\,.\,\vec{B}=\left|\vec{A}\times\vec{B}\right|$. Then the value of $\left|\vec{A}-\vec{B}\right|$ will be :

(A) $\sqrt{A^2+B^2+\sqrt{2}AB}$

(B) $\sqrt{A^2+B^2}$

(C) $\sqrt{A^2+B^2-\sqrt{2}AB}$

(D) $\sqrt{A^2+B^2+2AB}$

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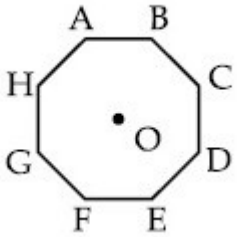
Q21. JEE Main 2021 (Online) 25th February Morning Shift

Open solution

In an octagon ABCDEFGH of equal side, what is the sum of

$\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD} + \overrightarrow{AE} + \overrightarrow{AF} + \overrightarrow{AG} + \overrightarrow{AH}$,

if, $\overrightarrow{AO} = 2\hat{i} + 3\hat{j} - 4\hat{k}$



(A) $-16\hat{i} - 24\hat{j} + 32\hat{k}$

(B) $16\hat{i} + 24\hat{j} - 32\hat{k}$

(C) $16\hat{i} + 24\hat{j} + 32\hat{k}$

(D) $16\hat{i} - 24\hat{j} + 32\hat{k}$

Q22. JEE Main 2019 (Online) 8th April Evening Slot

Open solution

Let $\left|\overrightarrow{A_1}\right| = 3$, $\left|\overrightarrow{A_2}\right| = 5$ and $\left|\overrightarrow{A_1} + \overrightarrow{A_2}\right| = 5$. The value of $\left(2\overrightarrow{A_1} + 3\overrightarrow{A_2}\right).\left(3\overrightarrow{A_1} - 2\overrightarrow{A_2}\right)$ is :-

(A) −118.5 (B) −112.5 (C) −99.5 (D) −106.5

Q23. JEE Main 2019 (Online) 10th January Evening Slot

Open solution

Two vectors $\overrightarrow{A}$ and $\overrightarrow{B}$ have equal magnitudes. The magnitude of $\left(\overrightarrow{A} + \overrightarrow{B}\right)$ is 'n' times the magnitude of $\left(\overrightarrow{A} - \overrightarrow{B}\right)$. The angle between $\overrightarrow{A}$ and $\overrightarrow{B}$ is -

(A) $\sin^{-1}\left[\frac{n-1}{n+1}\right]$

(B) $\sin^{-1}\left[\frac{n^2-1}{n^2+1}\right]$

(C) $\cos^{-1}\left[\frac{n^2-1}{n^2+1}\right]$

(D) $\cos^{-1}\left[\frac{n-1}{n+1}\right]$

Q24. JEE Main 2019 (Online) 10th January Evening Slot

Open solution

Two forces P and Q, of magnitude 2F and 3F, respectively, are at an angle θ with each other. If the force Q is doubled, then their resultant also gets doubled. Then, the angle θ is -

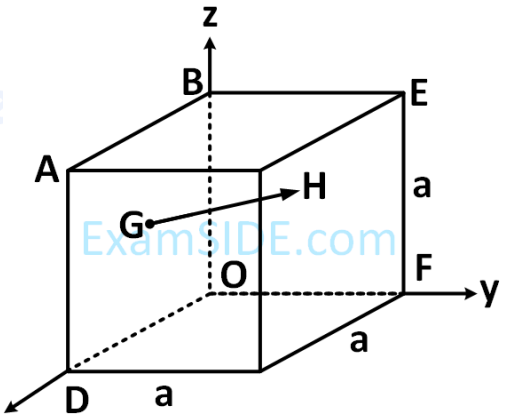
(A) 90° (B) 60° (C) 30° (D) 120°

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Q25. JEE Main 2019 (Online) 10th January Morning Slot

In the cube of side 'a' shown in the figure, the vector from the central point of the face ABOD to the central point of the face BEFO will be -

Open solution



(A) $\frac{1}{2}a\left(\hat{k}-\hat{i}\right)$

(B) $\frac{1}{2}a\left(\hat{j}-\hat{i}\right)$

(C) $\frac{1}{2}a\left(\hat{j}-\hat{k}\right)$

(D) $\frac{1}{2}a\left(\hat{i}-\hat{k}\right)$

Q26. JEE Main 2018 (Online) 16th April Morning Slot

Let $\vec{A}$ = $\left(\hat{i}+\hat{j}\right)$ and, $\vec{B}=\left(2\hat{i}-\hat{j}\right)$. The magnitude of a coplanar vector $\vec{C}$ such that $\vec{A}.\vec{C}=\vec{B}.\vec{C}=\vec{A}.\vec{B}$, is given by :

Open solution

(A) $\sqrt{\frac{10}{9}}$

(B) $\sqrt{\frac{5}{9}}$

(C) $\sqrt{\frac{20}{9}}$

(D) $\sqrt{\frac{9}{12}}$

Q27. JEE Main 2015 (Online) 11th April Morning Slot

A vector $\vec{A}$ is rotated by a small angle $\Delta\theta$ radian ($\Delta\theta$ < <1) to get a new vector $\vec{B}$. In that case $\left|\vec{B}-\vec{A}\right|$ is :

Open solution

(A) 0

(B) $\left|\vec{A}\right|\left(1-\frac{\Delta\theta^2}{2}\right)$

(C) $\left|\vec{A}\right|\Delta\theta$

(D) $\left|\vec{B}\right|\Delta\theta-\left|\vec{A}\right|$

Q28. AIEEE 2004

If $\vec{A}\times\vec{B}=\vec{B}\times\vec{A}$, then the angle between A and B is

Open solution

(A) $\frac{\pi}{2}$

(B) $\frac{\pi}{3}$

(C) π

(D) $\frac{\pi}{4}$

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Numerical 14 questions

Q29. JEE Main 2025 (Online) 23rd January Morning Shift

Two particles are located at equal distance from origin. The position vectors of those are represented by $\vec{A} = 2\hat{i} + 3n\hat{j} + 2\hat{k}$ and $\vec{B} = 2\hat{i} - 2\hat{j} + 4p\hat{k}$, respectively. If both the vectors are at right angle to each other, the value of n^{-1} is _____ .

Open solution

Q30. JEE Main 2024 (Online) 9th April Evening Shift

The resultant of two vectors $\vec{A}$ and $\vec{B}$ is perpendicular to $\vec{A}$ and its magnitude is half that of $\vec{B}$. The angle between vectors $\vec{A}$ and $\vec{B}$ is _____$^\circ$.

Open solution

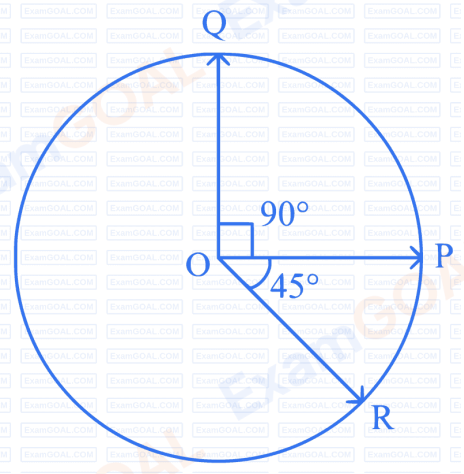
Q31. JEE Main 2024 (Online) 9th April Morning Shift

If $\vec{a}$ and $\vec{b}$ make an angle $\cos^{-1}\left(\frac{5}{9}\right)$ with each other, then $|\vec{a}+\vec{b}| = \sqrt{2}|\vec{a}-\vec{b}|$ for $|\vec{a}| = n|\vec{b}|$. The integer value of n is _____.

Open solution

Q32. JEE Main 2024 (Online) 8th April Morning Shift

Three vectors $\overrightarrow{\mathrm{OP}}, \overrightarrow{\mathrm{OQ}}$ and $\overrightarrow{\mathrm{OR}}$ each of magnitude A are acting as shown in figure. The resultant of the three vectors is $\mathrm{A}\sqrt{x}$. The value of x is _____.



Open solution

Q33. JEE Main 2024 (Online) 6th April Morning Shift

For three vectors $\vec{A} = (-x\hat{i} - 6\hat{j} - 2\hat{k})$, $\vec{B} = (-\hat{i} + 4\hat{j} + 3\hat{k})$ and $\vec{C} = (-8\hat{i} - \hat{j} + 3\hat{k})$, if $\vec{A} \cdot (\vec{B} \times \vec{C}) = 0$, then value of x is _____.

Open solution

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Q34. JEE Main 2024 (Online) 4th April Morning Shift

Two forces $\overline{\mathrm{F}}_1$ and $\overline{\mathrm{F}}_2$ are acting on a body. One force has magnitude thrice that of the other force and the resultant of the two forces is equal to the force of larger magnitude. The angle between $\vec{F}_1$ and $\vec{F}_2$ is $\cos^{-1}\left(\frac{1}{n}\right)$. The value of $|n|$ is _____.

Open solution

Q35. JEE Main 2024 (Online) 30th January Evening Shift

A vector has magnitude same as that of $\vec{A} = 3\hat{i} + 4\hat{j}$ and is parallel to $\vec{B} = 4\hat{i} + 3\hat{j}$. The x and y components of this vector in first quadrant are x and 3 respectively where $x =$ _____.

Open solution

Q36. JEE Main 2023 (Online) 25th January Morning Shift

If $\overrightarrow{P} = 3\hat{i} + \sqrt{3}\hat{j} + 2\hat{k}$ and $\overrightarrow{Q} = 4\hat{i} + \sqrt{3}\hat{j} + 2.5\hat{k}$ then, the unit vector in the direction of $\overrightarrow{P} \times \overrightarrow{Q}$ is $\frac{1}{x}\left(\sqrt{3}\hat{i} + \hat{j} - 2\sqrt{3}\hat{k}\right)$. The value of x is _____.

Open solution

Q37. JEE Main 2023 (Online) 24th January Morning Shift

Vectors $a\hat{i} + b\hat{j} + \hat{k}$ and $2\hat{i} - 3\hat{j} + 4\hat{k}$ are perpendicular to each other when $3a + 2b = 7$, the ratio of a to b is $\frac{x}{2}$. The value of x is _____.

Open solution

Q38. JEE Main 2022 (Online) 28th July Morning Shift

If the projection of $2\hat{i} + 4\hat{j} - 2\hat{k}$ on $\hat{i} + 2\hat{j} + \alpha\hat{k}$ is zero. Then, the value of α will be _____.

Open solution

Q39. JEE Main 2022 (Online) 26th July Evening Shift

If $\vec{A} = (2\hat{i} + 3\hat{j} - \hat{k})\,\mathrm{m}$ and $\vec{B} = (\hat{i} + 2\hat{j} + 2\hat{k})\,\mathrm{m}$. The magnitude of component of vector $\vec{A}$ along vector $\vec{B}$ will be _____ m.

Open solution

Q40. JEE Main 2021 (Online) 22th July Evening Shift

Three particles P, Q and R are moving along the vectors $\overrightarrow{A} = \hat{i} + \hat{j}$, $\overrightarrow{B} = \hat{j} + \hat{k}$ and $\overrightarrow{C} = -\hat{i} + \hat{j}$ respectively. They strike on a point and start to move in different directions. Now particle P is moving normal to the plane which contains vector $\overrightarrow{A}$ and $\overrightarrow{B}$. Similarly particle Q is moving normal to the plane which contains vector $\overrightarrow{A}$ and $\overrightarrow{C}$. The angle between the direction of motion of P and Q is $\cos^{-1}\left(\frac{1}{\sqrt{x}}\right)$. Then the value of x is _____.

Open solution

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Q41. JEE Main 2021 (Online) 25th February Evening Shift

If $\vec{P} \times \vec{Q} = \vec{Q} \times \vec{P}$, the angle between $\vec{P}$ and $\vec{Q}$ is $\theta(0^\circ < \theta < 360^\circ)$. The value of '$\theta$' will be _____$^\circ$.

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Q42. JEE Main 2020 (Online) 7th January Evening Slot

The sum of two forces $\vec{P}$ and $\vec{Q}$ is $\vec{R}$ such that $\left|\vec{R}\right| = \left|\vec{P}\right|$. The angle θ (in degrees) that the resultant of $2\vec{P}$ and $\vec{Q}$ will make with $\vec{Q}$ is .....

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Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **B**	3. **A**	4. **B**	5. **B**
6. **C**	7. **D**	8. **C**	9. **C**	10. **B**
11. **B**	12. **A**	13. **C**	14. **A**	15. **A**
16. **B**	17. **B**	18. **B**	19. **D**	20. **C**
21. **B**	22. **A**	23. **C**	24. **D**	25. **B**
26. **B**	27. **C**	28. **C**		

Numerical

29. **3**	30. **150**	31. **3**	32. **3**	33. **4**
34. **6**	35. **4**	36. **4**	37. **1**	38. **5**
39. **2**	40. **3**	41. **180**	42. **90**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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